

DU CHENG

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Summary

Senior Data Scientist with 8 years of experience bridging **AI research and production**, specializing in Applied AI and **Generative AI** across LLMs, NLP, recommendation systems, and computer vision. Experienced in translating ambiguous business challenges into scalable AI solutions through end-to-end machine learning development, cross-functional collaboration, and data-driven product innovation.

Education

University of Chicago

Master of Science in Computer Science

Chicago, IL

September 2019 – June 2020

University of North Carolina at Chapel Hill

Bachelor of Arts in Physics & History

Chapel Hill, NC

August 2013 – May 2017

Experience

Fidelity Investments

Boston, MA

Senior Data Scientist

January 2024 – Present

- Leading discovery and architecture design for an AI/ML-based margin disbursement automation initiative spanning 1.7M annual transactions and 22.5K+ hours of manual effort, defining a target system that integrates document classification, data extraction, and rules-based decisioning to enable intelligent routing and reduce manual review, and proposing an agentic system to iteratively improve decision rules through feedback-driven refinement, while aligning stakeholders on technical design and operational constraints.
- Architected an end-to-end LLM-assisted interpretability pipeline that converts propensity model outputs into personalized, human-readable explanations for campaign decision support, combining rule-based learning (BoolXAI), automated feature selection, and localized reasoning to operationalize model interpretability for downstream business use. Documented in a published research paper.
- Led the AI-Branch Workforce Management initiative, owning roadmap and requirement translation while developing an appointment likelihood model and engineering its integration with a constraint-based optimizer and GenAI-powered conversational interface to support interactive scheduling and real-time scenario analysis. Partnered closely with architects and engineers to define system design, align on implementation, and provide ongoing guidance throughout delivery. This multi-part system is deployed and piloting across 10 branches. The solution was a regional finalist in the 2025 Brokerage Hackathon.
- Developed a propensity-based lead generation model for an annual charitable giving campaign, prioritizing high-likelihood prospects across email and rep-led channels to improve targeting effectiveness and outreach strategy, delivering a 35% lift in account opening rate, up to 84% lift in lead conversion across segments, and \$126.5M in contributions from converted leads.
- Contributing to the open-source AI community, extending recommendation capabilities in widely used recommendation libraries and maintaining projects with 1M+ downloads.

Data Scientist

November 2021 – December 2023

- Served as the AI lead for the Fidelity Bloom mobile app, designing and implementing applied ML solutions for a mobile-first product by developing a multi-armed bandit framework for content card recommendations and a graph-based customer journey model to enable next-best-action personalization, transforming a previously non-AI application into a data-driven experience.
- Developed a data synthesis and augmentation pipeline for large-scale document processing, leveraging AWS Textract for upstream structured data extraction and generating high-quality training data to improve downstream image understanding models, including experiments with fine-tuning and prompt engineering across language and vision models, resulting in a 15% performance improvement.
- Developed novel self-learning pipeline with a language model and handwriting generator for Optical Character Recognition (OCR) to automate handwritten form processing, with results presented at ACM DocEng 2023.

Walgreens Boots Alliance

Chicago, IL

Data Scientist

July 2020 – July 2021

- Led customer segmentation and acquisition initiatives for Order Online, Pick Up In Store, developing supervised and unsupervised machine learning models to identify high-value customer segments across a 100M+ customer base.
- Designed and developed a customer audience management platform that streamlined audience selection for marketing campaigns, standardized campaign measurement workflows, and improved the scalability and efficiency of data science operations.

Lowe's Companies, Inc

Mooresville, NC

Data Scientist

June 2019 – August 2019

- Created a Lowe's network wide staffing tool that uses shift assignment algorithms and employment classification models, optimized store staffing efficiency and close rate, built a React/NodeJS application to replace the current vendor-supplied workforce management software, deployed to nationwide stores significantly improving staffing efficiency for management.

Junior Data Scientist

May 2017 – June 2019

- Partnered with business leaders to drive business insights into analytical solutions, analyzed and performed experiments with large datasets on distributed systems and cloud platforms, designed large-scale client facing web applications, worked with cross-functional teams to achieve key business goals.
- Optimized retail inventory by building an internal supply chain application, manipulated terabytes of omni-channel sales/supply data in Hive/Spark, simulated large scale supply chain processes in python/PySpark/C++.

Relevant Publications

- **Du Cheng**, Serdar Kadioğlu, and Xin Wang. BoolXLLM: LLM-Assisted Explainability for Boolean Models. **arXiv preprint**, 2026.
- Adrian Boteanu, **Du Cheng**, and Serdar Kadioğlu. Read-Write-Learn: Self-Learning for Handwriting Recognition. **ACM Symposium on Document Engineering (DocEng)**, 2023.
- **Du Cheng**, Doruk Kilitçioğlu, and Serdar Kadioğlu. Bias Mitigation in Recommender Systems to Improve Diversity. **CIKM Workshops**, 2022.

Technical Proficiency

Generative AI: LangChain, LangGraph, HuggingFace, GenAI Gateway, Prompt Engineering

Programming Languages: Python, SQL, Java, JavaScript

Machine Learning: Natural Language Processing (NLP), Recommendation Systems, Explainable AI, Computer Vision

Cloud & Data Platforms: Microsoft Azure, AWS, Databricks, Snowflake, Apache Spark

Certification

Product Management Certificate (Cornell University)

Building AI Agents and Agentic Workflows (IBM)

Machine Learning Engineer Path (Pluralsight)